Linear algebra: Exercise Chapter 6

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1. Compute determinant and inverse of the following matrices, if any.

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 6 & 0 & 2 \\ 1 & 2 & 0 \\ 4 & 1 & 1 \end{bmatrix}.$$

Answer.

Matrix A:

$$\det(A) = (3)(0) - (-1)(1) = 1,$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix}.$$

Matrix B: Since two rows (first and third) of B are the same, so $\det(B) = 0$, and therefore, matrix B is singular and does not have an inverse.

Matrix C:

$$\det(C) = 6\begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} - 0\begin{vmatrix} 1 & 0 \\ 4 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix} = 6 \cdot 2 + 2 \cdot (-7) = 12 - 14 = -2$$

Since $\det(C) \neq 0$, the matrix C is invertible.

To find the minors, M_{ij}, delete the row and column of each element in C, then compute the determinant of the submatrix. Next, compute cofactors $c_{ij} = (-1)^{i+j} M_{ij}$.

$$M_{11} = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2, \quad M_{12} = \begin{vmatrix} 1 & 0 \\ 4 & 1 \end{vmatrix} = 1, \quad M_{13} = \begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix} = -7,$$

$$M_{21} = \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} = -2, \quad M_{22} = \begin{vmatrix} 6 & 2 \\ 4 & 1 \end{vmatrix} = -2, \quad M_{23} = \begin{vmatrix} 6 & 0 \\ 4 & 1 \end{vmatrix} = 6,$$

$$M_{31} = \begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix} = -4, \quad M_{32} = \begin{vmatrix} 6 & 2 \\ 1 & 0 \end{vmatrix} = -2, \quad M_{33} = \begin{vmatrix} 6 & 0 \\ 1 & 2 \end{vmatrix} = 12.$$

$$\text{Cofactor}(C) = \begin{bmatrix} 2 & -1 & -7 \\ 2 & -2 & -6 \\ -4 & 2 & 12 \end{bmatrix}, \qquad \text{Adj}(C) = \text{Cofactor}(C)^T = \begin{bmatrix} 2 & 2 & -4 \\ -1 & -2 & 2 \\ -7 & -6 & 12 \end{bmatrix},$$

$$C^{-1} = \frac{1}{-2} \begin{bmatrix} 2 & 2 & -4 \\ -1 & -2 & 2 \\ -7 & -6 & 12 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 2 \\ \frac{1}{2} & 1 & -1 \\ \frac{7}{2} & 3 & -6 \end{bmatrix}.$$

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2. For an $n \times n$ matrix A with $\det(A) = -2$, find:

   (a) $\det(2A^T)$;

   (b) $\det(-A)$;

   (c) $\det(A^2)$;

   (d) $\det(A^{-1})$;

   (e) $\det(C)$.

Answer.

(a) $\det(2A^T) = 2^n \det(A^T) = 2^n \det(A) = -2^{n+1}$.

(b) $\det(-A) = (-1)^n \det(A) = (-1)^{n+1}2$.

(c) $\det(A^2) = \det(A.A) = \det(A)\det(A) = \det(A)^2 = 4$.

(d) $1 = \det(I_n) = \det(A.A^{-1}) = \det(A)\det(A^{-1}) \Longrightarrow \det(A^{-1}) = \frac{1}{\det(A)} = -\frac{1}{2}$.

(e) $\det(C) = \det(C^T) = \det(\det(A)A^{-1}) = \det(A)^n \det(A^{-1}) = \det(A)^n \frac{1}{\det(A)} = \det(A)^{n-1} = (-2)^{n-1}$.

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3. Determine the determinant of the following matrices by using cofactors and selecting an appropriate row/column.

$$A = \begin{bmatrix} 8 & -11 & 7 \\ -1 & 0 & 4 \\ 7 & -3 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 3 & 0 & 3 & 0 \\ 4 & 8 & 1 & 1 \\ 7 & 0 & 1 & 0 \end{bmatrix}.$$

Answer.

Matrix A: Expand along the second row,

$$\det(A) = -(-1)\begin{vmatrix} -11 & 7 \\ -3 & 2 \end{vmatrix} + 0 - 4\begin{vmatrix} 8 & -11 \\ 7 & -3 \end{vmatrix} = \big(-22-(-21)\big) - 4\big(-24-(-77)\big) = -1 - 4(53) = -213.$$

Matrix B: Expand along the second column,

$$\det(B) = -8\begin{vmatrix} 1 & 1 & 2 \\ 3 & 3 & 0 \\ 7 & 1 & 0 \end{vmatrix} = -(8)(2)\begin{vmatrix} 3 & 3 \\ 7 & 1 \end{vmatrix} = -16(3-21) = 288.$$

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4. Find the determinant of the following matrices by using determinant rules (don't compute it by using cofactors). You can use the results from the previous problem.

$$A = \begin{bmatrix} 4 & -1 & 7 \\ -9 & 2 & 2 \\ 4 & -1 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -1 & 0 \\ -9 & 2 & 0 \\ 14 & 7 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 3 & 9 \\ -1 & 0 & 4 \\ 0 & 3 & 13 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad E = \begin{bmatrix} 6 & 2 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 11 \end{bmatrix},$$

$$F = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 7 & 0 & -2 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 11 & 0 & 0 & 0 \\ 0 & 0 & 0 & 10 \\ 0 & 0 & -4 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & -11 & 7 \\ -1 & 0 & 4 \\ 1 & -9 & 6 \end{bmatrix}.$$

Which of the above matrices are invertible?

Answer.

Matrix A: The first row and the third row of the matrix A are identical, so $\det(A) = 0$.

Matrix B: The third column of the matrix B is zeros vector, so $\det(B) = 0$.

Matrix C: $r_1 + r_2 = r_3$, so $\det(C) = 0$.

Matrix D: It is a diagonal matrix. $\det(D) = 5(-6)(-1) = 30$.

Matrix E: It is a triangular matrix. $\det(E) = 6 \times 1 \times 11 = 66$.

Matrix F: It is a triangular matrix. $\det(F) = 4 \times 9 \times (-2) = -72$.

Matrix G:

$$\det(G) = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 11 & 0 & 0 & 0 \\ 0 & 0 & 0 & 10 \\ 0 & 0 & -4 & 0 \end{vmatrix} \overset{r_1 \leftrightarrow r_2}{=} -\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 11 & 0 & 0 \\ 0 & 0 & 0 & 10 \\ 0 & 0 & -4 & 0 \end{vmatrix} \overset{r_3 \leftrightarrow r_4}{=} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 11 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & -4 \end{vmatrix} = -440.$$

Matrix H:

$$\det(H) = \begin{vmatrix} 1 & -11 & 7 \\ -1 & 0 & 4 \\ 1 & -9 & 6 \end{vmatrix} \overset{r_2 \to r_2 + r_1}{=} \begin{vmatrix} 1 & -11 & 7 \\ 0 & -11 & 11 \\ 1 & -9 & 6 \end{vmatrix} \overset{r_3 \to r_3 - r_1}{=} \begin{vmatrix} 1 & -11 & 7 \\ 0 & -11 & 11 \\ 0 & 2 & -1 \end{vmatrix} \overset{r_3 \to r_3 + \frac{2}{11} r_2}{=} \begin{vmatrix} 1 & -11 & 7 \\ 0 & -11 & 11 \\ 0 & 0 & 1 \end{vmatrix} = -11.$$

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5. Solve the following two systems of equations by using Cramer's rule.

$$a) \begin{cases} 12x + 3y = 15, \\ 2x - 3y = 13. \end{cases} \qquad b) \begin{cases} x + y - z = 6, \\ 3x - 2y + z = -5, \\ x + 3y - 2z = 14. \end{cases}$$

Answer.

Part (a):

$$A = \begin{bmatrix} 12 & 3 \\ 2 & -3 \end{bmatrix}, \qquad \det(A) = \begin{vmatrix} 12 & 3 \\ 2 & -3 \end{vmatrix} = -36 - 6 = -42,$$

$$A_1 = \begin{bmatrix} 15 & 3 \\ 13 & -3 \end{bmatrix}, \qquad \det(A_2) = \begin{vmatrix} 15 & 3 \\ 13 & -3 \end{vmatrix} = -45 - 39 = -84,$$

$$A_2 = \begin{bmatrix} 12 & 15 \\ 2 & 13 \end{bmatrix}, \qquad \det(A_2) = \begin{vmatrix} 12 & 15 \\ 2 & 13 \end{vmatrix} = 156 - 30 = 126.$$

$$x = \frac{\det(A_x)}{\det(A)} = \frac{-84}{-42} = 2, \qquad y = \frac{\det(A_y)}{\det(A)} = \frac{126}{-42} = -3.$$

Part (b):

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & -2 & 1 \\ 1 & 3 & -2 \end{bmatrix}, \quad \det(A) = 1\begin{vmatrix} -2 & 1 \\ 3 & -2 \end{vmatrix} - 1\begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} + (-1)\begin{vmatrix} 3 & -2 \\ 1 & 3 \end{vmatrix} = (4-3) - (-6-1) - (9-(-2)) = -3,$$

$$A_1 = \begin{bmatrix} 6 & 1 & -1 \\ -5 & -2 & 1 \\ 14 & 3 & -2 \end{bmatrix}, \quad \det(A_1) = 6\begin{vmatrix} -2 & 1 \\ 3 & -2 \end{vmatrix} - 1\begin{vmatrix} -5 & 1 \\ 14 & -2 \end{vmatrix} + (-1)\begin{vmatrix} -5 & -2 \\ 14 & 3 \end{vmatrix} = 6(4-3) - (10-14) - (-15-(-28)) = -3,$$

$$A_2 = \begin{bmatrix} 1 & 6 & -1 \\ 3 & -5 & 1 \\ 1 & 14 & -2 \end{bmatrix}, \quad \det(A_2) = 1\begin{vmatrix} -5 & 1 \\ 14 & -2 \end{vmatrix} - 6\begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} + (-1)\begin{vmatrix} 3 & -5 \\ 1 & 14 \end{vmatrix} = (10-14) - 6(-6-1) - (42-(-5)) = -9,$$

$$A_3 = \begin{bmatrix} 1 & 1 & 6 \\ 3 & -2 & -5 \\ 1 & 3 & 14 \end{bmatrix}, \quad \det(A_3) = 1\begin{vmatrix} -2 & -5 \\ 3 & 14 \end{vmatrix} - 1\begin{vmatrix} 3 & -5 \\ 1 & 14 \end{vmatrix} + 6\begin{vmatrix} 3 & -2 \\ 1 & 3 \end{vmatrix} = (-28-(-15)) - (42-(-5)) + 6(9-(-2)) = 6,$$

$$x = \frac{\det(A_1)}{\det(A)} = \frac{-3}{-3} = 1, \qquad y = \frac{\det(A_2)}{\det(A)} = \frac{-9}{-3} = 3, \qquad z = \frac{\det(A_3)}{\det(A)} = \frac{6}{-3} = -2.$$

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6. Let n be an odd number and A an $n \times n$ matrix such that $A^T = -A$. Show that A is not invertible.

Answer.

$\det(A) = \det(A^T) = \det(-A) = (-1)^n \det(A) = -\det(A) \Longrightarrow \det(A) = 0.$

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7. Suppose $\det(A) = 1$ and you know all the cofactors of A. How can you find A?

Answer.

Using the cofactors of A, we can form the cofactor matrix C. Then, since $\det(A) = 1$, we have

$$A^{-1} = \frac{C^T}{\det(A)} = C^T \Longleftrightarrow A = (C^T)^{-1}.$$

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8. Write a Python code that prompts the user to enter a matrix and then compute the determinant of that.

Answer.

```python
import numpy as np

def recursive_determinant(A):
    """
    Compute the determinant of a square matrix A recursively.
    """
    n = A.shape[0]

    # Base case for 1x1 matrix
    if n == 1:
        return A[0, 0]

    # Base case for 2x2 matrix
    if n == 2:
        return A[0, 0]*A[1, 1] - A[0, 1]*A[1, 0]

    # Recursive case for n > 2
    det_value = 0
    for col in range(n):
        sub_matrix = np.delete(np.delete(A, 0, axis=0), col, axis=1)
        cofactor = ((-1) ** (col)) * A[0, col]
        det_value += cofactor * recursive_determinant(sub_matrix)
    return det_value


# ------------------ MAIN PROGRAM ------------------

# Input from user
matrix_str = input("Enter the square matrix A (e.g. [[1,2,3],[4,5,6],[7,8,9]]): ")
A = np.array(eval(matrix_str), dtype=float)

print("A =")
print(A)

# Check if matrix is square
rows, cols = A.shape
if rows != cols:
    raise ValueError("Matrix is not square. Cannot compute determinant.")

# Compute determinant
det_A = recursive_determinant(A)
print(f"det(A) = {det_A:.2f}")
```